

A Comprehensive Review on the Medicinal Plants for the Treatment of Headache

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ABSTRACT

Mother nature cure is the trademark sort of recovering treatment to fix diverse medical issues. Headache is started by hack, wheezing, stressing, fever, or rest; there are uncommon signs on appraisal and neurological or visual side effects. Headache is a constant indication in everybody and is a famous issue in youth and pubescence, all things considered in adulthood. A broad hunt of logical information banks, for example, Elsevier, PubMed, Medline and Scopus, has been practiced to discover results identifying with the calming, against nociceptive, and pain-relieving impacts of signified restorative herbs. We present a few herbs which their headache impacts have been assessed in clinical and trial examines. White Willow Bark, Latjira, Guduchi, Ashwagandha, Coriander, Brahmi, Nagarmotha, Jatamansi, and Imli, are some of the introduced medicinal herbs in this review. Since the treatment of cerebral pain is anything but a one-dimensional cure, this article attempts to arrive at a multidimensional restorative way to deal with headache with the assistance of natural medication and change in way of life.

Keywords: Headache, Herbs, Natural product, Nociceptive, Prostaglandins

INTRODUCTION

Humankind has utilized therapeutic plants to mend sicknesses and calm actual experiencing the foremost punctual occasions. As indicated by WHO (World Health Organization), 80% of populations of the developing nations depend upon conventional medicament, generally, it is plant source drugs. Therapeutic plants are the wellsprings of essential medical services for the duration of the planet for several years. Presently the pattern is changing and individuals favor the restorative plants as they contain naturalistic products which are successful, synthetically adjusted and have fewer toxic effects when contrasted to artificial chemicals [1].

Herbal remedy is the characteristic type of recuperating treatment to fix different health problems. Right now, natural medicaments are back into conspicuousness because of the manufactured medicament. It has been acknowledged that nature has the most appropriate responses to all the illnesses influencing the human body now and again. When the manufactured medicines neglect to be potent or indicate genuine toxic results, it is the plant medication that brings facilitates [2]. It is very much demonstrated that spices have phenomenal properties for treating pain and nervousness influencing the focal sensory system in nearly a similar way as some doctors prescribed medications, without the toxic effects.

All over the world, medicines of plant origin have been doing amazements in healing different infections, including mental recuperating [3]. All the medicinal plant has covered in this paper is generally secure and effective in daily uses [4].

Headache is a continuous symptom in everyone and is an eminent issue in youth and puberty, for what it's worth in adulthood. Headache is originated by cough, sneezing, straining, fever, or sleep; there are unusual signs on assessment and neurological or visual symptoms. An abundance or absence of sleep could cause a headache [5]. Headache is a general and worldwide symptom with a complicated and miscellaneous set of origin. The headache itself could be hurting and disabling aspects that cause substantial levels of incapacity [6].

The brain is not responsive to pain as a result of its paucities receptors of pain. Several zones of the neck and head do have a sense of pain due to receptors of pain. These embody extracranial arteries, giant veins, portions of the brain-stem, meninges of middle meningeal artery, falx cerebri, sinuses receive blood from the veins, cranial nerves, spinal nerves, muscles of head-and-neck, teeth, the lining of the mouth, ears, and eyes. The Pial arteries, instead of the Pial veins are area units accountable for the production of pain and cause headache [7].

Headache typically is preceded by acute upper tract infection then the patient develops a fever,

unilateral severe pain, and nasal obstruction. The international society of headache (ISH) has classified headache, initial primary headache like migraine, headache, cluster headache, and trigeminal nerve cephalgia, or secondary headache that is rare, like a medical specialty sickness, vascular pathology, infection or injuries of cranial nerves. The tension type headache is the most typical variety of headache that 78% of the world's population is suffering from it. The headache has markedly worsened throughout the overuse of medicines like aspirin [8]. In kids more youthful than 15 years, aspirin isn't suggested due to the worry of Reye's syndrome [5].

Inhibitions of PGE₂ synthesis will result in enhanced sodium desorption, inflicting peripheral lump that is the most typical nephritic impact of NSAIDs. For acute and/or treatment of headache, the foremost often reported facet effects of NSAIDs area unit gastrointestinal (stomach and gut) symptoms, such as Gas, abdomen irritation and ulcers, abdomen pain, nausea, vomiting, diarrhea/constipation, heartburn, gastrointestinal (GI) bleeding and nephrotoxicity (fig. 1). The nose and throat examination of headache embody tissue layer lump, nasal congestion, and infected nasal discharge [8]. All sorts of headache area unit coded as ICD-9 code 784[9].

Arachidonic acid is derived from phospholipids (lipids of a membrane). Metabolism of arachidonic acid is done by prostaglandin synthases (PGA₂), which, through its cyclooxygenase (COX) and endoperoxides actions, outcomes in the process and production of PGG₂ to PGH₂. Prostacyclins (PGI₂), PGD₂, PGE₂, PGF₂, and TXA₂ are derived from PGH₂ by prostaglandin synthases. PGE₂ is chiefly present in the Brain, Kidney, Vascular smooth muscle, and Platelets. PGE₂ is proceeding as a vasodilator by reacting on vascular smooth muscle to cause blood vessel dilation. This result is increased blood circulation. This includes high blood pressure and headache pain (fig. 2) [10-13].

Table 1 is showing the different mechanisms of herbal plants through which they give analgesic effect in headache. Ongoing examinations demonstrate PG synthases are a significant helpful objective for pain and inflammation, change over PG precursors into organically dynamic prostanoids, plays a critical role in pain. PGE₂ synthases have gotten a lot of ongoing consideration contrasted with the others. Among the PGE receptors, EP₁ is the primary receptor answerable for mucosal security. Prostanoids, their receptors, and terminal prostaglandin (PG) E synthases, especially microsomal PGE synthase-1 (mPGES-1) alongside cyclooxygenase-2 (COX-2)

proteins are exceedingly significant parts of pain sensitization in the central sensory system and peripheral sensory system. After COX inhibitors were undesired results, for example, ulcerogenic impacts and cardiovascular risk of COX-2 specific agents may emerge because of the adjustment of thromboxane and PGI₂ (Prostacyclins). Arachidonic acid delivered from phospholipids by phospholipases A₂ is a substrate for cyclooxygenase-2 (COX-2), producing prostaglandin G₂ (PGG₂) and afterward PGH₂. PGH₂ is the substrate for most terminal prostaglandin synthases; PGs have both supportive of and mitigating properties. Prostaglandin E (EP) receptors have been demonstrated to be confined in trigeminal ganglion neurons where the stimulation brings about CGRP discharge [12] (fig. 3).

MEDICINAL PLANTS POSSESSING ANALGESIC ACTIVITY

White Willow Bark (*Salix alba* L., Salicaceae):

Willow bark (fig. 4) is one of only a handful few plant crude materials that contain characteristic active substances that have analgesic, anti-inflammatory, fever-reducing, and anti-rheumatic effects [14]. The most well-known species utilized restoratively are *Salix alba* L., *Salix purpurea* L., *Salix daphnoides* Vill., and *Salix nigra* Marshall. These families of barks contain higher quantity of salicylate precursors (1.5 to 15%) [15, 16].

White willow bark has analgesic and anti-inflammatory property, due chiefly to salicin content. The parts of willow species that can be utilized are chiefly represented by bark [17, 18].

The word salicin is gotten from the Latin name for the willow, *Salix*. Salicin is the white unpleasant-tasting powder [19]. Salicin contains 23–26% all-out salicin derivatives, which are metabolized in vivo to salicylic acid [31]. The white willow bark hinders the building structure of inflammatory moderators, for example, cyclooxygenase (COX-1, COX-2) intervened prostaglandins (PGs), leukotrienes (LTs), and cytokines (a large group of proteins, glycoproteins or peptides which are released by cells of the immune system) [20].

In oral administration, those active compounds (salicylic glycosides, mostly salicin and salicortin) are hydrolyzed by Gut flora in the upper digestive tract into salicyl alcohol which is assimilated and oxidized to salicylic acid. White Willow bark doesn't make harm stomach mucous membrane [21].

The therapeutic properties of willow bark can be ascribed to the presence of salicylic glycosides (SGs), chiefly salicin, and salicortin. [14]. Willow bark is especially plentiful in SGs, which furnish

the significant plant resource with analgesic, anti-inflammatory, antipyretic and anti-rheumatic properties and convey comparable impacts to manufactured acetylsalicylic acid, one of the most broadly produced drugs in the world [17]. Willow is regularly used as potent an analgesic, because of an Aspirin like active compound or salicylic acid which is present in the bark of the willow tree [22].

The bark extract of White willow is broadly utilized for a situation related to pain, headache, inflammation, fever, osteoarthritis, acute back pain, menstrual cramps, antiseptic, generalized pain, and flu symptoms with fever [23]. Manufacture aspirin (acetylsalicylic acid) hydrolyzed to salicylic acid and acetic anhydride which promotes the acetylation of cell macromolecules in the stomach mucous membrane and prompts their harm. Active compounds from willow bark are absorbed in more slow- rate than manufactured aspirin, and they apply longer enduring healing or restorative effects [22, 24, 25].

No genuine unfavorable impacts were accounted for from willow bark extracts delivering 120–240 mg salicin daily for up to 8 weeks [26, 16].

The willow bark of salicin at a dose of 120 to 240 mg gave analgesic effect similar to a 500 mg Aspirin due to attendance of flavonoids and polyphenol in bark extract [23].

Latjira (*Achyranthes aspera*, *Amaranthaceae*):

Achyranthes aspera (fig. 5) is one of the plants utilized for therapeutic purposes [27]. It is a yearly, stiff herb, about 0.3 to 0.9m high, and is a typical plant found in badlands, all through roadside and rural fields. It is known by various names in India like Latjira and Chirchira in Hindi, Apamargah, Chirchitaa, and Shikhari in Sanskrit, Chirchitaa in Unani, Pricklychaff flower plant in English, Nayuruvi in Tamil, Uttaraene in Telugu, Kutri in Punjabi, and Kadalad in Malayalam. [28, 29].

There are over 10 cultivars accessible. Popular species incorporate *A. aspera*, *A. aspera* late ovata, *A. aspera* porphyristachya, *A. aspera* villosior, *A. aspera* velutina, *A. aspera* borbonica, *A. aspera* nigro olivacea, *A. aspera* pubescens and *A. aspera* rubrofusca [30].

Flowers of *A. aspera* have greenish and pinkish petals creating rice seeds. The significant synthetic constituents are alkaloids, flavonoids, saponins, sugars, proteins are available. Seeds contain saponin A and B. *A. aspera* seed of saponin A was recognized as D-glucuronic acid and saponins B was distinguished as a β -D-galactopyranosyl ester of D-glucuronic acid. Leucine, isoleucine, phenylalanine, valine, oleanolic acid glycoside are active constituents

present in the seed part of *A. aspera*. Achyranthine, linalool, oleanolic acid, arachidic, behenic, betaine, hentriacontane, ecdysterone, and achyranthes saponins A, B, are the significant substance constituents found in *A. aspera*. The seeds powder of *A. aspera* gives calmness from solidness and headache. The leaves and seeds of *Achyranthes aspera* show pain-relieving activity [2, 30, 32, 33].

Headache is caused by mainly inflammation in brain stem, Latjira has been recommended to be a possible lead for another kind of anti-inflammatory agent having a double inhibitory action on phospholipase A (corticosteroid) and cyclooxygenase (COX-1, COX-2) which is responsible for pain and inflammation [34].

Ethanollic extract of leaf of *A. aspera* (400 mg/kg) has produced an analgesic effect at 30, 60, 90 and 120min using Tail flick response and Hot plate methods. The leaf extract produce an analgesic effect by reduction the formalin induced pain at dose of 400 mg/kg in albino mice using Formalin-induced pain test [35].

The plant is utilized as analgesic, anti-inflammatory, antiviral, toothache, gastric tonic, nasal contamination, carminative, diuretic, natural blood purifier, expectorant, an antidote for snake. The entire plant is utilized in gaseous distention, gastritis, cough, bronchitis, colds, asthma, piles, urinary bladder stones, rheumatism, scabies, and hypoglycemic, liver and kidney diseases, muscles sustaining, allergy, cardiovascular, cholera, rabies, colic, gonorrhea, pneumonia, hydrophobia, and skin diseases [27, 29, 32, 36-38].

Guduchi (*Tinospora cordifolia*, *Menispermaceae*):

Guduchi (fig. 6) is a natural plant and has promising properties as a propitious plant [39]. Its territory ranges across a wide area in India spreading from south to north [40]. *T. cordifolia* has various native names in different Indian dialects like that Amrita and Gilo in Kashmiri, Gila in Punjabi, Giloe, and Gurcha in Hindi, *Tinospora* in English, Chakralakshanika in Sanskrit, Galac and Gila in Gujarati, Gulvel in Marathi, Gulancha in Bengali, Sittamrytu, and Chittamrutu in Malayalam, Guluchi in Oriya, Somida in Telugu and Sindal in Tamil [40, 43].

Pain, inflammation, and Fever are generally treated with *T. cordifolia* [41]. Anti-inflammatory effects of *T. cordifolia* were diminished synthesis of proinflammatory cytokines, for instance, IL-1, IL-6, and TNF- α [42].

T. cordifolia contains glycosides, alkaloids, steroids, flavonoids, and terpenoids in the elevated piece of the plant. In this way, the noticed pain-relieving movement might be

ascribed to any of these phytoconstituents. There are reports of the pain-relieving action of flavonoids which is hindering the building structure of prostaglandins [44] and also, substances like bradykinin, Substance P (powerful arbiters of pain) are delivered during inflammation. Anti-inflammatory agents like NSAIDs are utilized as analgesics and reduce pain by diminishing different arbiters of inflammation particularly PG. Gastritis is a habitual result of NSAIDs while *T. cordifolia* has gastro defensive movement which is an advantageous property while utilizing it as an analgesic [40] as it demonstrated pain-relieving action in both the systems so its pain-relieving movement may include focal (Narcotic receptors) just as peripheral (hindrance of PG synthesis) [45]. The Anti-inflammatory impact of *T. cordifolia* was accounted for to be more than aspirin (acetylsalicylic acid 100 mg/kg) [46].

The creators have proposed that the anti-inflammatory effect of *T. cordifolia* was because of obstruction with histamine and serotonin release [40]. The major phytoconstituent found in *T. cordifolia* is tinosporaside, tinosporidine, tinosporine, tinosporide, Isocolumbin, columbin, cordifolia, cardifol, heptacasanol, palmatine, diterpenoid furano lactone, furano diterpone, 6-sitosterol, clerodane [47].

T. cordifolia extract (300 mg/kg body weight) was discovered to possess an analgesic effect when compared to standard Pentazocine (10 mg/kg dose) using Abdominal writhing method and Hot plate method in rats [45].

The aqueous extract of stem of *T. cordifolia* exhibited a potent reduction in carrageenan-induced acute inflammation in rats. The aqueous extract of *T. cordifolia* root has also been discovered to inhibit paw edema induced by carrageenan in rats [40].

T. cordifolia has demonstrated many promising organic exercises, for example, analgesics or pain relieving, anti-inflammatory or calming and against pyretic activities [18, 45, 48], general tonic, hostile to spasmodic [44], cancer prevention agent, heart-defensive, neuroprotective, antianxiety [49,50] osteoprotective, hepatoprotective, antidiarrheal, antihyperglycemic and antistress impacts. Guduchi is a rich wellspring of protein and micronutrients, for example, iron, manganese, copper, zinc, calcium, and phosphorus. It additionally contains numerous auxiliary plant metabolites, for example, flavonoids, glycosides, steroids, alkaloids, and terpenes [43].

Jatamansi (Nardostachys jatamansi, Valerianaceae):

N. jatamansi (fig. 7) is an imperative Phyto plant with various cures. It is the fundamental plant of Ayurvedic material medicament [51]. The main part of *N. jatamansi* of leaves and rhizomes are utilized for headache, fever, cold, cough, and sickness [52-54]. It is found in India (Himachal Pradesh, Uttarakhand, Sikkim, and Arunachal Pradesh) [55].

N. jatamansi has different local names in India, Jatamansi, Bal-scorch, Balachadi, Chhar, Kalichada, Sumbul In Hindi, Indian spikenard, Indian valerian, Muskroot, True spikenard in English, Jatamansi, Bhytajata, Bhutakesi, Mishi, Pisita, Tapasvini in Sanskrit, Balchhad, Sunbuluttee in Urdu, butijatt, Kuklipot in Kashmiri, Jatamansi in Bengali, Jatamanshi in Marathi, Jatamansi, Kalichhad in Gujarati, Kannad in Malayalam Jatamanshi in Telugu [27, 56].

N. jatamansi rhizomes are broadly abused for the formulation of medicines because of the presence of naturally dynamic phytochemicals, for example, coumarins, alkaloids, lignans, neolignans, terpenes, sesquiterpenes, sesquiterpenoids, steroids, sterols, tannins, glue, flavonoids, sugars, gums, and glycosides and their subordinates, with a wide scope of organic properties [57, 58]. Various significant medicinal compounds additionally present *N. jatamansi* incorporate actinidine (alkaloid), jatamansin, jatamansinone, jatamansinol (coumarins), jatamol A and jatamol B (sesquiterpenoids), nardostachone, valeranone, nardosinone [59, 60].

Jatamansi (terpenic coumarins), Angelicin (furocoumarins), Beta-patchoulene (tricyclic sesquiterpene), and Beta-sitosterol (phytosterol) compounds of root and rhizome part of *N. jatamansi* have a anti-inflammatory activity [16].

N. jatamansi has been appeared to obstruct the synthesis of PGE₂, interleukin (IL-1 and IL-6), and tumor necrosis factor (TNF- α). Interleukin-6 assumes a double job both as an anti-inflammatory as well as a pro-inflammatory cytokine [61, 62]. *Nardostachys jatamansi* builds levels of the accompanying key synapses in the cerebrum, Gamma-aminobutyric acid (GABA), norepinephrine, Dopamine, Serotonin [56, 63].

N. jatamansi has a history of utilization in Ayurveda and conventional medication. *N. jatamansi* of various plant portions of concentrate (root, stem, leaves, blossom, and bulb) have been utilized for different treatments[64] like that tonic for headache, cold, cough, fever, skin illnesses, dysentery[65], treat stress, migraine, hypertension, cardiovascular diseases[66, 67], blood purification, gastritis, wounds, jaundice, stomach issues, heart problems[68], chronic

fever, intestinal worms, food poisoning, sickness [52], memory enhancement [53, 69], liver issue, jaundice, kidney problems, and insomnia [70].

The different number of *N. jatamansi* formulated products like Patanjali Divya Medhavati and Dimag Paushtik Rasayan Tablet, Stressoneel Capsule, Sadamanjil Chooranam Powder, Remigin Nasal drops Drop, and Tengin pushpadi Tailam Oil which is utilized to treat cold, wheezing, runny nose, cough, fever, insomnia, hypertension, migraine, stress, despondency, and tension which are straightforwardly related with headache [55]. *N. Jatamansi* will be a significant plant to convey research for different sicknesses. Ethanol extract of root of *N. jatamansi* was discovered to be non-poisonous up to the dose of 200mg/kg body weight and did not cause any passing, accordingly it is considered as protected [51].

Brahmi (*Bacopa monnieri*, *Schrophulariaceae*):

Brahmi (fig. 8) is gotten from the word "Brahma" the legendary maker of the Hindu pantheon of Indian religion. The cerebrum is the focal point of imaginative movement in the human body, those components that improve mental wellbeing is called Brahmi. *B. monnieri* is a little, crawling herb with various branches, little leaves, and light purple blossoms. In India, it grows normally in wet soil, shallow water, bogs [2, 72], grows 6 crawls to 3 feet in tallness and the whole plant alongside the roots can be utilized restoratively [71].

It is known by different local names in India like Brahmi, Jalneem, Bama, and Mandukaparni in Hindi, Aindri, Sarasvati, Jalamimba in Sanskrit, Water hyssop, Thyme leaved gratiola, Monnier's fence hyssop in English, Bama, Jalnam, Brahmi in Marathi, Brahmi in Urdu, Sambrani-aku, Sambarani, Sambarenu in Telugu [2, 27, 73].

B. monnieri has analgesic, anti-inflammatory, and antipyretic activities [27, 72, 74-78]. *B. monnieri* has the capacity to restrain the production of cyclooxygenase (COX-2), Lipoxygenase (LOX) [79-82], and PGE2 initiating inflammation which causes headache [85]. It likewise represses the arrival of pro-inflammatory mediators (PGE2, histamine, serotonin, bradykinin) [83-85] and TNF- α activities [79, 80, 83]. This action might be because of the essence of the triterpenoids and bacosides in it [72].

Viable and significant medicinal chemical compounds incorporate Saponins (Bacoside A, bacoside B, bacopasaponins, monnierin, and D-mannitol), Alkaloids (Herpestine, brahmine, and hydrocotyline), Glycosides (Thanakunide and asiaticoside), Flavonoids (Apigenin, luteolin-7-glucoside, and luteonin), and Phytochemicals

(Betulinic acid, Betulic acid, β -sitosterol, and stigmasterol) [27, 77, 85-87, 90].

It was concluded that at a dose of 250 and 500 mg/kg body weight of methanolic extract of *B. monnieri* showed an analgesic activity using Acetic acid induced writhing test when compared to Diclofenac Sodium [91]. Methanolic and aqueous extracts of *B. monnieri* showed potent anti-inflammatory activity at 100 mg/kg body weight dose in rats using Carrageenan induced paw edema method. The extracts of hexane and petroleum ether did not show anti-inflammatory activity when compared to a 25 mg of Aspirin [92].

In Ayurveda, Brahmi is considered as a memory enhancer, nervine-tonic, neuroprotective role, insomnia, antidepressant activity, anti-ulcerogenic activity, adaptogenic activity, blood pressure, broncho vasodilatory activity, gastrointestinal, alzheimer's disease, anti-anxiety activity, antibacterial activity, rheumatism, anti-cancer activity, sedative, anticonvulsant activity, anticholinesterase activity, immunomodulatory activity, diuretic, alopecia, leprosy, eczema, anti-epileptic activity, antileishmanial activity, antimicrobial activity, antioxidant activity, antiparkinson, antispasmodic activity, antitubercular activity, anxiolytic activity, attention deficit disorder, tranquilizing activity, wound healing activity, enlargement of spleen, snake bite, endocrine effects, free radical scavenging, and cell stabilization activity [72-78, 85, 93, 94, 96-98].

Nagarmotha (*Cyperus rotundus*, *Cyperaceae*):

Cyperus Rotundus (fig. 9) has been classified as "the world's most exceedingly terrible weed. The plant of *C. rotundus* requires sun and damp conditions. It fills in sandy soil, just as in loamy wet fields and tropical rainforests [99]. It is a perpetual bush that achieves tallness of up to 40 cm, it has a dull greenish stem and the leaves are long and sharp, with a width of 1/6 to 1/3 inch. While the blossom stem has a three-sided cross-segment, the bloom is 2 to 8 inches long [100].

C. rotundus has a various vernacular names in India, Nagarmoth, Mutha, Motha, Korehi-jhar in Hindi, Mustaka, Ghana, Mutha, Musta in Sanskrit, Nut-grass, Purple nut-grass, Coco grass, Nut sedge in English, SadKufi, Nagarmotha in Urdu, Nagarmotha, , Moth in Gujarati, Bimbal, Moth, Nagarmotha in Marathi, Tungamustalu in Telugu [27, 73].

C. rotundus is broadly utilized as analgesic, anti-inflammatory, and antipyretic in another medication around the globe [101-104]. In the component of *C. rotundus*, the anti-inflammatory property of α -cyperone is an associate with

inhibition of cyclooxygenase (COX-2), and interleukin (IL-6) [105].

Phytochemical study of *C. rotundus* have demonstrated that the significant compounds are flavonoids, terpenoids, mono- and sesquiterpenes, cyprene, cyperene, α -copaene, α -cyperone, caffeic corrosive, coumarin, ellagic acid, salicylic acid, linolenic acid, oleic acid, linoleic acid, oleanolic acid, myristic acid, catechin, quercetin, genistein, and rutin [101, 102, 106-110].

As per Ayurveda, it is utilized as antirheumatic, anticonvulsant activity, antibiotic, antifungal, antiemetic activity, anti dysenteric, antimalarial, anti-microbial, diuretic, antioxidant activity, improves circulation, anticancer activity, gastroprotective activity, promotes memory, alopecia, anodyne, dyslepsia, antiparasitic, hemodialysis, sedative, stimulant, indigestion, antispasmodic, tranquilizer, vasodilator, anthelmintic, astringent, hypoglycemic agent, infectious diarrheal agent, urinary infections, skin diseases, cervical cancer and menstrual disorders [2, 27, 73, 97-102, 110].

Ashwagandha (*Withania somnifera*, **Solanaceae**):

Withania somnifera (fig. 10) is an esteemed herb in Ayurvedic medication, and as such was utilized and developed for quite a long time in India (Uttar Pradesh, Madhya Pradesh, Gujarat, Rajasthan, and Punjab). *W. somnifera* have 1.5-m high shrub with ovate leaves and yellowish-green blossoms. The leaves of *W. somnifera* are 10 cm long and 2.5-5 cm wide. The Roots of *W. somnifera* are 20-30 cm long and 6-12 mm. It has a trademark scent, taste severe and horse-like smell [2, 27].

W. somnifera is known by different nearby names in India like Rasbhari, Asgandh, Punir in Hindi, Ashvagandha, Varaha, Turangi, Turagi-gandha in Sanskrit, Indian ginseng, Winter cherry, Rennet in English, Asgandh nagori in Urdu, Ghoda, Ghodakun in Gujarati, Aksan in Punjabi, Ashwagandha in Bengali, Amkulang, Amukkura in Tamil [4, 27, 111].

The extract of roots contains biologically active chemical constituents are Alkaloids (Withanine, withanane, withasomnine, somnine, somniferine, nicotine, isopelletierine, tropeltigloate, anaferine, and somniferinine), Flavonoids (Quercetin, kaempferol), Steroidal Lactones (Withaferin-A, withanone, withanolides), Steroids (stigmasterol, β -sitosterol, cholesterol, diosgenin, stigmastadien, sitoindosides), and Nitrogen containing Somnitol somnisol, and withanol [27, 111-115].

W. somnifera has analgesic, anti-inflammatory, and antipyretic [114, 116]. Powder of ashwagandha roots was found to have a strong

inhibitory effect on, and other inflammatory markers such as cytokines including interleukin (IL - 6), tumor necrosis factor (TNF- α) and nephritis [117, 118].

Active constituents of *W. somnifera* Withaferin A exhibited an analgesic activity at dose of 30 mg/kg body weight in mice when compared to standard drug Indomethacin using Acetic acid induced writhing method [114].

In Ayurveda, Ashwagandha is considered as nervine tonic, memory enhancer, anti-stress, anxiety, cold, cough, bipolar disorder, antioxidant, immunomodulatory, hemoprotective, anti-tumor, antimalarial potential, antimicrobial, cardioprotective, hypocholesteremia, neuroprotective, adaptogenic, antidepressant, anxiolytic, anticancer, alzheimer's disease, parkinson's disease, epilepsy, and diuretic [27, 111-115, 119].

Coriander (*Coriandrum sativum* Linn, **Apiaceae**):

Coriandrum Sativum (fig. 11) is a yearly, herbaceous plant. It has thin, axle formed roots, an erect tail, substitute green leaves, and little, pinkish-white flowers [120, 121]. The color of the developed seeds is yellow brown [122].

C. Sativum has a different vernacular name in India, Dhania, Dhanya in Hindi, Kusthumbari, Dhanayaka in Sanskrit, Coriander Collender, Chinese parsley, Cilantro in English, Dhana, Dhanis in Gujarati, Dhanya, Dhana, Kothmir Khotbir in Marathi, Dhania in Urdu, Dhania in Punjabi, Bhoti, Dhana, Dhania, in Bengali, Dhaniya, Kottamalli in Tamil [123, 124]. In India, it is primarily found in Rajasthan, Madhya Pradesh, Karnataka, Andhra Pradesh, Bihar, and Tamil Nadu [125].

Various substance constituents, for example, volatile constituents, flavonoids (luteolin, kaempferol, and quercetin), phenolic acids (gallic acid, caffeic acid, and ferulic acid), alkaloids, tannins, steroid, saponins, isocoumarins, and coriandrone. Isocoumarins, coriandrone A, B, C, D, and E were isolated from the aerial parts of *C. sativum* together with two known isocoumarins, coriandrin, and dihydrocoriandrin. Protocatechinic acid, caffeic acid, and glycitin were characterized as the major polyphenolics of coriander [125-131].

The other oil constituents were camphor, linalool, geranyl acetate α -pinene, γ -terpinene, limonene, geraniol, linalyl acetate, phellandrene, and limonene were the chief components of coriander seed essential oil [131, 132].

C. sativum is used in headache [2, 4, 123, 133]. *C. Sativum* seed extracts and essential oils have analgesic and anti-inflammatory activities [134-137]. It has been accounted for that linalool

(acyclic monoterpenes alcohol) and geranyl acetate which is commonly found as major constituents of coriander oil, play a major role in the analgesic activity [4, 130].

The possible mechanism of *C. sativum* inhibits the synthesis of prostaglandins (PGE₂), leukotrienes, and antioxidant activity. The analgesic effect of the linalool having *C. sativum* is mediated through inhibition of central pain receptors (including opioid, glutamatergic, glutamate, and adenosinergic receptors) [138-141].

Aqueous extract of *C. sativum* seeds exhibited an analgesic activity at dose of 200 mg/kg body weight in mice when compared to a 5 mg of Morphine sulphate using Hot plate method. Ethanolic extract of *C. sativum* fruits exhibited anti-inflammatory activity at doses of 50 to 200 mg/kg body weight in rats when compared to a 5mg of Diclofenac sodium using Carrageenan induced paw edema method [132].

In the toxicological study it is shown that the active constituent of *C. sativum* linalool is not toxic at maternal doses of 500 to 1000 mg/kg body weight in day [132].

In Ayurveda, coriander is used as antioxidant, sedative-hypnotic, anticonvulsant, memory-enhancer, anti-ulcer, diuretic, sub-acid gastritis, cholesterol-lowering, gut modulatory, blood pressure-lowering, anxiolytic, hepatoprotective, diarrhea, anti-diabetic, vomiting, anti-mutagenic, anti-protozoal, relief of insomnia, anxiety, antibacterial, antifungal, antioxidant, respiratory system, urinary system, diaphoretic, carminative, stimulant, insecticidal, sedative, hypnotic activity, anxiolytic activity, anorexia, protective role against lead toxicity, antifungal, anti-feeding activity, anticancer, heavy metal detoxification, dyspepsia, and anthelmintic [122-124, 128, 129, 142-147]. *C. Sativum* is used as traditional medicine, with no poisonous impacts, and especially its principal constituent, linalool, can be considered as generally safe and effective [148].

Sweet Basil (*Ocimum basilicum*, Lamiaceae):

In Ayurveda, *Ocimum basilicum* (fig. 12) is known as Sweet Basil. It is an annual, aromatic, herb, and perennial plants [149, 150]. It is known by various names in India such as Babui tulsi, Subzah, Jangli tulsi in Hindi, Vishva-Tulasi, Varavara in Sanskrit, Sweet or Common Basil, Basil in English, Nasabo, Damaro in Gujarati, Nazbo, Babi tulsi in Marathi and Tirunitrupachchai, Karandai in Tamil [27, 151, 152].

O. basilicum has a significant active constituent such as flavonoids, glycoside, triterpenoids, phenolic compounds, proteins, amino acids, tannins, gums, steroids, flavones, and saponins [153]. The whole plant of *O. basilicum* contains

linoleic acid, caffeic acid, vanillic acid, δ -gurjunene, apigenin, menthol, camphor, rosmarinic acids, lauric acid, quercetin, α -linolenic, rutin, myristic acid, acid, carpic acid, arachidonic acid, 1,8-cineole, limonene, menthone, menthol, camphor, chicoric acid, estragol, eucalyptol, ocimene, menthone, cyclohexanone, cyclohexanol, and myrcenol [153-156].

Linalool (monoterpene alcohol), Eugenol (phenylpropanoid), Bornyl acetate (Acetate ester), Germacrene (sesquiterpene), α -bisabolol (sesquiterpene alcohol) have analgesic and anti-inflammatory activities [157-160] by inhibition of the biosynthesis of pain mediators, such as prostaglandins (PGE₂), prostacyclins and opioid receptors [161]. *O. basilicum* extract also inhibits the proinflammatory cytokines and mediators (TNF- α , Interleukin-1 β , and Interleukin-2, which responsible for inflammation [162]. *O. basilicum* shows analgesics activity when compared with standard drug, Aspirin [163].

The result shows that *O. basilicum* oil (200 mg/kg body weight) and fixed oil (0.1 ml/kg body weight) demonstrated anti-inflammatory activity in rats using animal hind paw edema model. *O. basilicum* oil possesses potent analgesic effect by inhibition of cyclooxygenase (COX) and lipoxygenase (LO) pathway of arachidonic acid metabolism without accountable toxic effects [156].

O. basilicum is used as a whole herb to treat a headache. It is also used in colds, coughs, infections, pyrexia, migraine, aids digestion, stomach aches, constipation, nephritis, hepatotoxicity, and gout [126, 152-155].

Imli (*Tamarindus indica*, Fabaceae):

Imli (fig. 13) is one of the tree type plants with active medicinal purposes [164]. It is amazingly discovered all over India. *Tamarindus indica*'s fruits are additionally found fundamentally summer season. The seed of *T. indica* is tanish dark in color. The fruit pulp of *T. indica* has an acidic taste due to the high contents of tartaric acid. [165].

T. indica is count in dicotyledonous plants [166]. It fills well in both semi-parched and damp rainstorm atmospheres and can develop on a wide scope of soil types of field. It is a tree in the jungles, it can endure temperatures up to 47°C, however, is extremely touchy to frost. It is grown all-state in India aside from the Himalayas and western dry regions [167].

Tamarindus indica has a different name in India, Ambli, Imli in Hindi, Ambia, Tintiri, Amalika in Sanskrit, Imli, Imbli in Punjabi, Imli in Urdu, Tamarind, Indian date in English, Omlika, Tentuli, Koina in Odiya, Ambali in Marathi, Amlī,

Ambli in Gujarati, Tinturi, Tentul, Ambli, in Bengali, Amilam, Pulie, Ambilam in Tamil [168, 169].

The root, bark, leaf, and pulp part of *T. indica* has a biologically active compound such as flavonoids, tannins, saponins, α -kaloids, carbohydrates, and phenol [170, 171]. Apigenin, catechin, limonene, β -amyrin, β -sitosterol, campesterol, naringenin, palmitic acid, linoleic acid, oleic acid, n-hexacosane, procyanidin B2, citric acid, tartaric acid, acetic acid, eicosanoic acid, succinic acid, taxifolin, eriodictyol, formic acid, malic acid [172-175].

T. indica has an analgesic and anti-inflammatory effects [176-178]. *T. indica* having flavonoids and tannins play an important role in an analgesic and anti-inflammatory effect due to inhibiting arachidonic acid lipoxygenation, synthesis of prostaglandins (PGE₂) and pro-inflammatory cytokines (IL-1, and TNF- α) [170, 179-181]. Catechins, quercetin apigenin, silymarin, and hydroxyethylrutinosides are flavonoids with dominant analgesic and anti-inflammatory properties by inhibition of building block structure of prostaglandin (PGE₂) and pro-inflammatory mediators [170, 171, 182]. The extract of *T. indica* has valuable analgesic and anti-inflammatory properties by inhibiting inflammatory mediators such as serotonin (5HT), histamine, leukotrienes (LT), and bradykinin [183].

It was shown by the results that the ethanol extract of *T. indica* seeds at doses of 25 to 50 mg/kg body weight produced the potent analgesic and anti-inflammatory effects by diminished the intensity of pro-inflammatory cytokines in Wistar rats. The methanolic extract of seeds at dose of 200 mg/kg body weight shows central analgesic and anti-inflammatory effects in Wistar rats. The aqueous extract of *T. indica* roots at doses of 300 to 600 mg/kg body weight increased inhibition of pain in Wistar rats. The ethanolic extract of *T. indica* leaves at dose of 400 mg/kg body weight exhibited analgesic action in Swiss albino mice. The extract exhibited analgesic and anti-inflammatory effects when compared to a 25mg of Diclofenac sodium by Tail immersion and Carrageenan induced paw edema methods [182].

Methanol extract of *T. indica* was evaluated for analgesic activity at dose of 200 mg/kg and 400 mg/kg body weight and anti-inflammatory activity at dose of 100 mg/kg, 200 mg/kg, and 400 mg/kg body weight using carrageenan-induced paw edema and tail immersion methods in rats. The extract exhibited an analgesic and anti-inflammatory effects when compared to standard Diclofenac sodium [178].

Ethyl acetate extract of *T. indica* seed exhibited maximum analgesic activity at 100 mg/kg body weight in 4 hour and 9 mg/kg of Diclofenac at 3 hour in rats by pain induced hot plate method. At dose of 50mg/kg, 100 mg/kg and 200 mg/kg of petroleum ether and ethyl acetate extracts of *T. indica* inhibited carrageenan-induced paw edema in rats using Carrageenan-induced hind paw edema method when compared to a 10 mg of Indomethacin [180].

Analgesic activity of root extract of *T. indica* was observed at a dose of 200 mg/kg body weight in rats using Hot plate method, and Acetic acid induced writhing method when compared to a 100 mg of aspirin [184].

In the toxicological study shows that the aqueous and ethanolic extract of *T. indica* leaves and pulp are atoxic and secure at doses of 1000 to 5000 mg/kg body weight in Wistar rats [182].

A whole plant of *T. indica* is used in headache, pyrexia, stomach ache, hepatotoxicity [185-187], and has antioxidant properties [188]. *T. indica* improves the absorption rate of analgesic and anti-inflammatory drugs like aspirin and ibuprofen [189]. The dose of plants extracts and chemical structure of active constituents of herbal plants is summarized in Table 2 and Table 3, respectively.

CONCLUSION

The outcomes of this present study give proof of relief in headache using all the medicinal plant. The different analgesic and anti-inflammatory drugs, for instance, non-steroidal anti-inflammatory drugs and narcotics present within the market initiate different results like a peptic ulcer, nephrotoxicity, and hepatotoxicity. Its utilization within the treatment of hurting situations connected headaches and others with some notable medical benefits. All the medicinal plants are a rich wellspring of elementary alkaloids, flavonoids, tannins, glycosides, triterpenoids, bacosides, and saponins.

Priority has been given during this paper that it's for the treatment of headaches. Herbal plants are common natural NSAIDs with minor or no toxic effects within the body. There is a requirement for additional investigation of isolation, purification, and formulation of the herbal extracts of dynamic constituents for analgesic activity.

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Not applicable

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No

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TABLE 1: MECHANISM OF HERBAL PLANTS

Common Name	Botanical Name	Family	Active Constituents	Mechanism of Action	Reference
White Willow bark	<i>Salix alba</i> L.	Salicaceae	Salicylic glycosides, salicin, and salicortin	Inhibition of prostaglandins (PGE_2), cyclooxygenase (COX-2)	14, 17, 18, 20, 21
Latjira	<i>Achyranthes aspera</i>	Amaranthaceae	Linalool, Oleanolic acid glycosides, amino acids, flavonoids, and saponins	Inhibits phospholipase A, cyclooxygenase (COX-1, COX-2)	27, 30, 32, 33, 34
Guduchi	<i>Tinospora cordifolia</i>	Menispermaceae	Tinosporine, isocolumbin, palmatine, flavonoids, terpenoids, glycosides, and alkaloids	Down-regulation of prostaglandins (PGE_2), and proinflammatory cytokines (IL-1b, IL-6, TNF- α)	40, 42, 44, 47
Jatamansi	<i>Nardostachys jatamansi</i>	Valerianaceae	Jatanmansi, β -patchoulene, Angelicin, alkaloids, tannins, flavonoids	Inhibits prostaglandins (PGE_2), proinflammatory cytokines (IL-1, IL-6, TNF- α), and serotonin	55, 56, 61, 62, 63
Brahmi	<i>Bacopa monnieri</i>	Schrophulariaceae	Apigenin, luteonin, Herpestine, saponins, glycosides, triterpenoids and bacosides	Restrain the production of cyclooxygenase (COX-2), prostaglandins (PGE_2), and pro-inflammatory mediators histamine, serotonin, bradykinin)	27, 72, 79, 80, 81
Nagarmotha	<i>Cyperus rotundus</i>	Cyperaceae	α -cyperone, salicylic acid, catechin, quercetin,	Inhibition of cyclooxygenase (COX-2), and interleukin (IL-6)	99, 101, 102, 104, 105
Ashwagandha	<i>Withania somnifera</i>	Solanaceae	Withaferin A, withanine,	Inhibition of interleukin (IL-6) and tumor necrosis	27, 111, 115, 117,

			quercetin, kaempferol, alkaloids and flavonoids	factor (TNF- α)	118
Coriander	Coriandrum Sativum L.	Apiaceae	Linalool, geranyl acetate, isocoumarins, caffeic acid, alkaloids, tannins and flavonoids	Inhibition of prostaglandins (PGE ₂) and opioid receptors	4, 138, 139, 140
Sweet Basil	Ocimum basilicum	Lamiaceae	Linalool, Eugenol, Bornyl acetate, Germacrene, α -bisabolol, apigenin, caffeic acid	Inhibition of pain mediators such as prostaglandins(PGE ₂), proinflammatory cytokines and mediators, and opioid receptors	157, 158, 159, 161, 160
Imli	Tamarindus indica	Fabaceae	Catechins, taxifolin, quercetin, apigenin, silymarin, luteolin, procyanidins, naringenin and hydroxyethylrutosides, tannins and alkaloids	Down-regulation of arachidonic acid lipoygenation, prostaglandins and pro-inflammatory mediators	170, 171, 180, 181, 182, 183

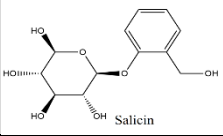
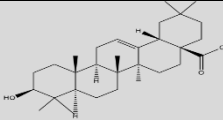
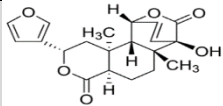
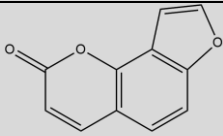
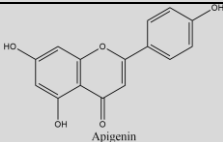
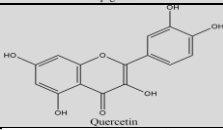
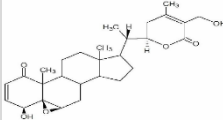
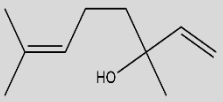
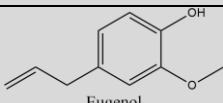
TABLE 2: A DOSE OF PLANTS EXTRACTS

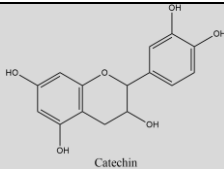
Species	Family	Plant parts	Solvent used in extraction	Doses	Reference
Salix alba L.	Salicaceae	Barks	Ethanol, and methanol extraction	120-240 mg/kg in Rats	14, 23, 95
Achyranthes aspera	Amaranthaceae	Whole plants	Methanol, Ethanol, and benzene extraction	400 mg/kg in Albino mice	34, 35, 89

<i>Tinospora cordifolia</i>	Menispermaceae	Stem, and whole plants	Aqueous, and ethanol extraction	300 mg/kg in Rats	40, 45, 88
<i>Nardostachys jatamansi</i>	Valerianaceae	Roots, and rhizomes	Ethanol, and methanol extraction	150, and 300 mg/kg in Albino rats	51, 55
<i>Bacopa monnieri</i>	Schrophulariaceae	Whole plants	Methanol, ethanol, and aqueous extraction	100, 250, and 500 mg/kg in Rats	78, 92
<i>Cyperus rotundus</i>	Cyperaceae	Rhizomes	Ethanol, and n-hexane extraction	-	105
<i>Withania somnifera</i>	Solanaceae	Roots	Aqueous extraction	30 mg/kg in Mice	111, 114
<i>Coriandrum Sativum</i> L.	Apiaceae	Seeds, fruits, and whole plants	Ethanol, and aqueous, extraction	50-200, 200 mg/kg in Mice and rats	132, 144
<i>Ocimum basilicum</i>	Lamiaceae	Whole plants	Methanol extraction	200 mg/kg in Rats	154, 161

Tamarindus indica	Fabaceae	Whole plants	Ethanol, methanol, ethyl acetate, petroleum ether extraction	25, 50, 100, 200, 300, 400, and 600 mg/kg in Wistar rats	178, 180, 182, 184
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TABLE 3: CHEMICAL STRUCTURE OF ACTIVE CONSTITUENTS OF HERBAL PLANTS

Plant Name	Chemical Structure	Reference
Salix alba L.	 Salicin	14, 21
Achyranthes aspera	 Oleanolic acid	30, 32, 33
Tinospora cordifolia	 Isocolumbin	27, 47
Nardostachys jatamansi	 Angelicin	16, 59, 60
Bacopa monnieri	 Apigenin	77, 85-87
Cyprus rotundus	 Quercetin	106-110
Withania somnifera	 Withaferin A	114, 115
Coriandrum sativum	 Linalool	2, 132, 133
Ocimum basilicum	 Eugenol	154, 157-160

Tamarindus indica	 Catechin	182, 189
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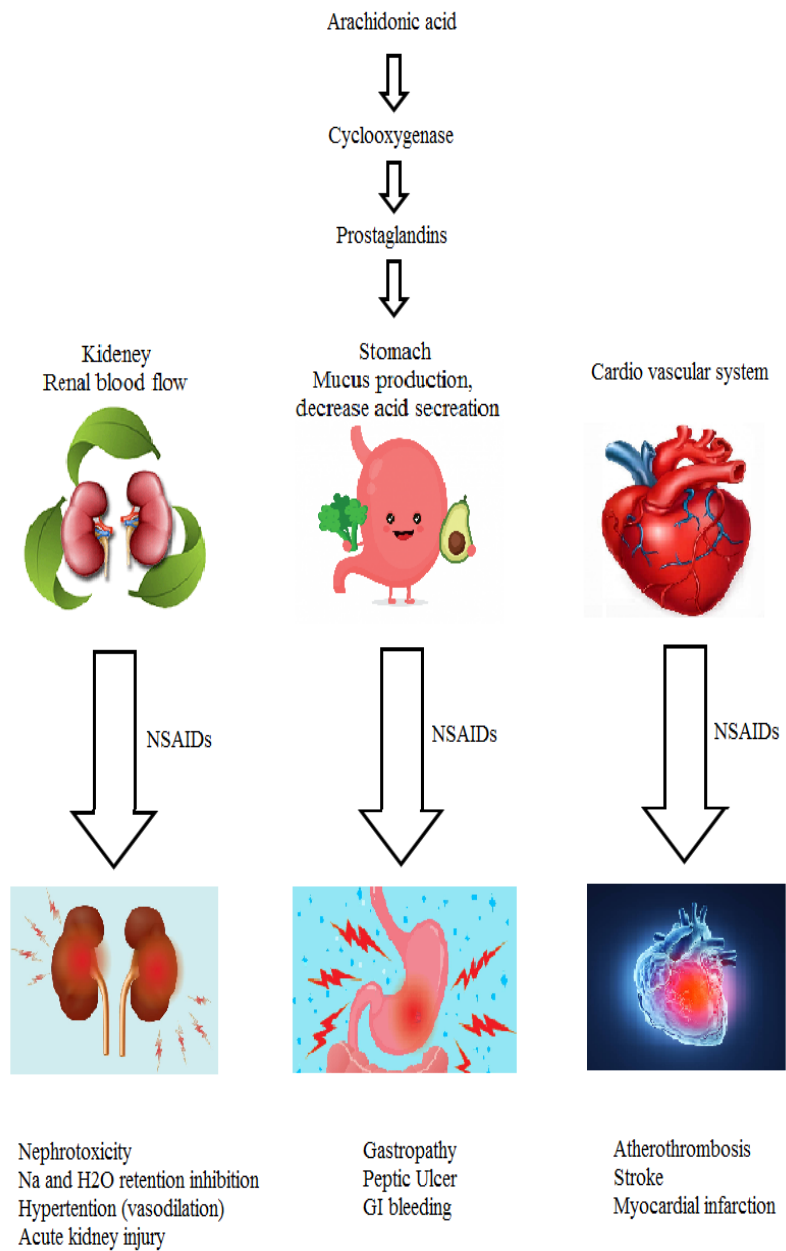


Fig. 1: Side effect of NSAIDs [10,12,191]

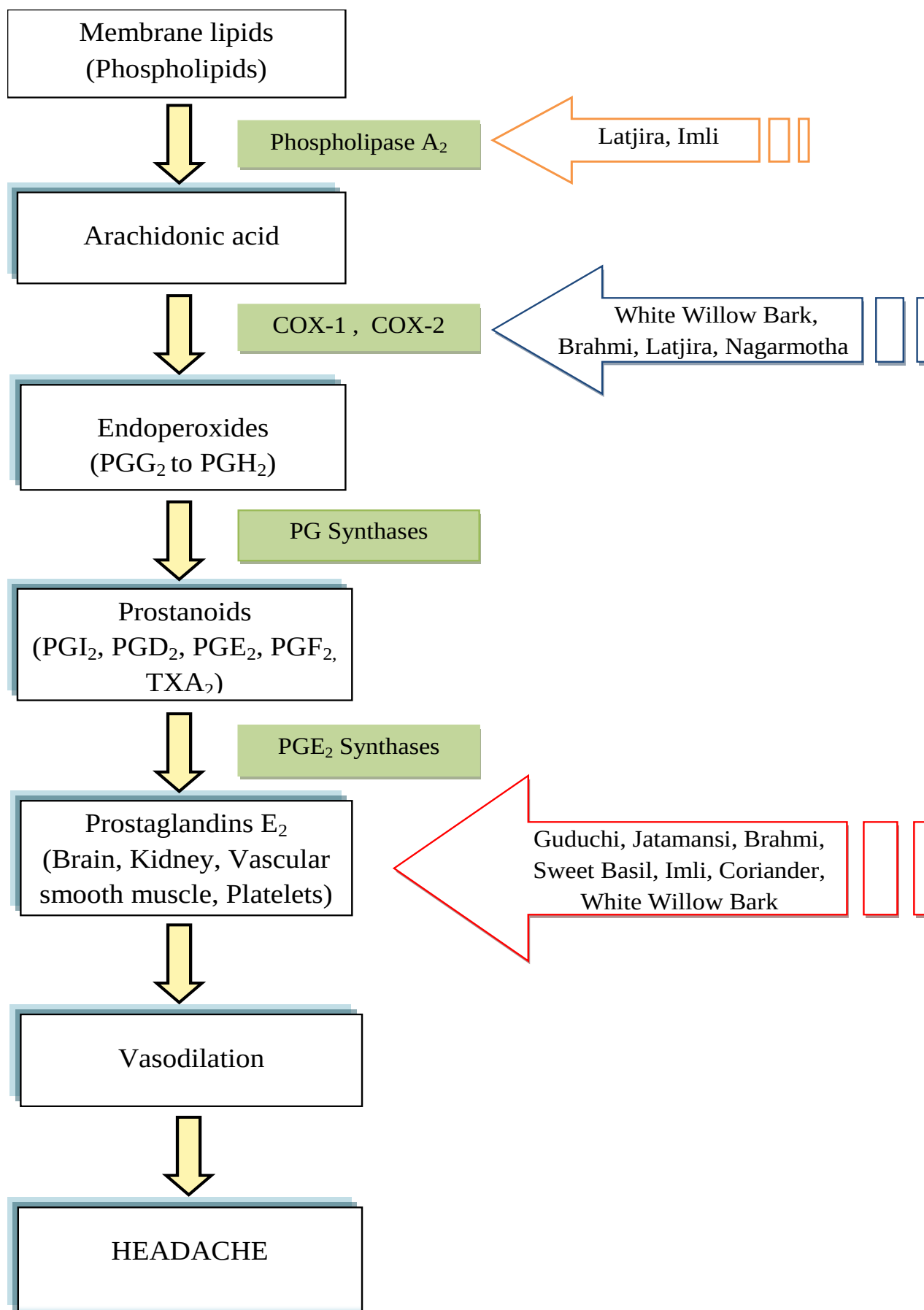


Fig.2: Mechanism of headache [10-13]

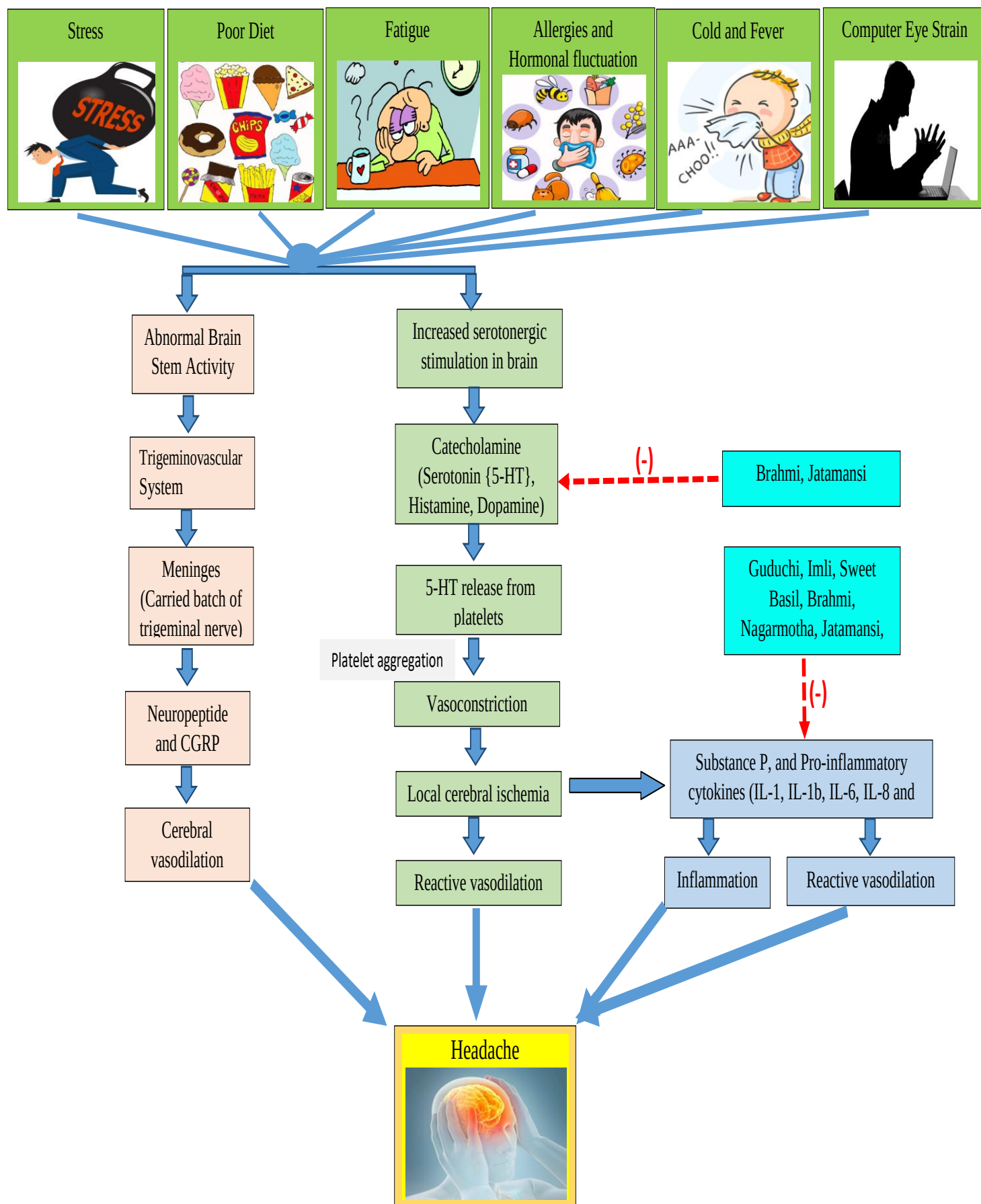


Fig. 3: Possible pathogenesis of headache [7,11,12,171,190]



Fig.4: Willow bark



Fig. 5: Latjira



Fig. 6: Guduchi



Fig. 7: Jatamansi



Fig. 8: Brahmi



Fig. 9. Nagarmotha



Fig. 10: Ashwagandha



Fig. 11: Coriander



Fig. 12: Sweet Basil



Fig. 13: Imli

Table and figure titles and legends:

Table 1: MECHANISM OF HERBAL PLANTS

Table 2: A DOSE OF PLANTS EXTRACTS

Table 3: CHEMICAL STRUCTURE OF ACTIVE CONSTITUENTS OF HERBAL PLANTS

Fig. 1: Side effect of NSAIDs

Fig. 2: Mechanism of headache

Fig. 3: Possible pathogenesis of headache

Fig. 4: Willow bark

Fig. 5: Latjira

Fig. 6: Guduchi

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